

GORIN, Russian Federation

WMO: 314890

Lat: 51.1931N

Lon: 136.6500E

Elev: 92

StdP: 100.23

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.2	-34.8	-40.2	0.1	-36.9	-37.8	0.1	-34.5	6.9	-11.8	5.5	-12.3	0.2	210	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.1	21.8	28.2	20.7	26.4	19.8	23.3	28.2	22.0	26.8	20.9	25.2	1.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.6	16.5	26.2	20.4	15.2	24.7	19.4	14.3	23.5	70.0	28.2	65.3	26.9	61.2	25.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.5	3.9	-39.7	33.2	2.0	1.8	-41.2	34.5	-42.4	35.6	-43.5	36.6	-45.0	37.9		
(5)			-39.8	24.9	2.0	1.1	-41.3	25.7	-42.4	26.4	-43.5	27.0	-44.9	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-24.1	-19.3	-9.4	1.7	10.2	16.0	19.9	18.4	11.9	2.6	-11.6	-22.2
	DBStd	16.36	6.06	5.40	5.40	4.75	4.47	3.98	3.54	2.93	3.79	4.15	7.11	6.22
	HDD10.0	4690	1058	820	603	252	54	3	0	1	23	231	647	999
	HDD18.3	6966	1316	1053	861	498	252	91	24	35	194	487	897	1257
	CDD10.0	900	0	0	0	4	62	182	308	261	81	2	0	0
	CDD18.3	134	0	0	0	0	1	21	74	37	2	0	0	0
	CDH23.3	1291	0	0	0	3	52	286	651	276	22	0	0	0
	CDH26.7	348	0	0	0	0	8	68	223	48	1	0	0	0
Wind	WSAvg	1.4	1.0	1.2	1.7	1.9	1.8	1.3	1.1	1.1	1.2	1.5	1.4	1.2
Precipitation	PrecAvg	640	18	14	21	44	64	66	101	110	68	54	30	29
	PrecMax	837	69	64	49	202	138	131	184	270	162	144	102	62
	PrecMin	360	2	3	2	5	6	16	0	19	16	9	2	1
	PrecStd	127	16	13	13	42	35	25	49	66	35	37	25	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.3	-2.8	7.2	21.7	27.7	31.1	33.7	30.0	25.9	18.5	6.7	-3.9
		MCWB	-5.0	-5.2	1.9	11.4	16.5	21.7	24.2	22.3	18.3	11.1	2.3	-4.7
	2%	DB	-8.2	-6.1	3.9	16.9	24.5	28.7	31.2	28.0	23.3	14.2	3.0	-6.7
		MCWB	-9.2	-7.5	-0.3	8.5	13.9	19.9	22.8	21.1	17.0	8.0	0.3	-7.5
	5%	DB	-11.6	-8.2	1.6	13.5	22.3	26.7	29.1	26.4	21.1	11.6	0.4	-9.6
		MCWB	-12.3	-9.4	-1.6	6.4	12.7	18.7	22.1	20.6	15.6	6.5	-1.5	-10.3
	10%	DB	-14.4	-10.1	-0.7	10.3	19.7	24.7	27.1	24.6	18.8	9.4	-1.4	-12.5
		MCWB	-15.0	-11.2	-3.2	4.6	11.6	17.7	21.1	19.6	14.2	5.4	-2.9	-13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.0	-5.0	2.0	12.4	17.4	22.5	26.2	24.1	19.6	11.6	2.6	-4.5
		MCDB	-4.0	-3.2	6.1	20.8	26.2	29.0	30.3	27.7	24.5	17.1	5.7	-4.0
	2%	WB	-9.1	-7.5	0.1	9.1	15.1	21.0	24.0	22.7	17.9	9.2	0.6	-7.5
		MCDB	-8.3	-6.1	3.5	16.4	22.1	27.3	29.6	26.6	22.2	12.5	2.6	-6.8
	5%	WB	-12.3	-9.4	-1.4	6.7	13.6	19.8	22.7	21.4	16.4	7.2	-1.2	-10.4
		MCDB	-11.6	-8.3	1.3	12.9	20.7	25.3	28.0	25.3	19.9	10.7	0.3	-9.6
	10%	WB	-15.0	-11.1	-3.1	4.7	12.2	18.4	21.6	20.2	15.0	5.7	-2.9	-13.1
		MCDB	-14.4	-10.0	-0.8	10.0	17.9	23.1	26.1	23.6	17.9	8.7	-1.4	-12.5
Mean Daily Temperature Range	5% DB	MDBR	10.3	13.0	12.3	10.6	12.1	11.7	10.1	9.9	11.1	10.0	9.8	9.3
		MCDBR	8.3	12.6	13.9	16.8	18.3	16.7	14.3	13.7	15.0	14.3	10.3	8.0
		MCWBR	7.9	11.1	10.2	9.3	9.0	8.5	6.8	7.5	8.8	8.6	7.8	7.6
	5% WB	MCDBR	8.1	12.2	12.6	16.1	16.2	13.9	12.7	11.8	13.5	13.3	9.0	7.7
		MCWBR	7.7	10.6	9.2	9.2	8.4	7.6	6.8	7.3	8.5	8.7	7.1	7.3
Clear-Sky Solar Irradiance	taub	0.269	0.303	0.375	0.486	0.565	0.500	0.497	0.423	0.381	0.392	0.328	0.271	
	taud	2.112	2.045	1.886	1.800	1.759	2.023	2.082	2.308	2.350	2.122	2.080	2.094	
	Ebn at Noon	730	800	794	746	706	766	761	807	802	684	629	652	
	Edh at Noon	76	109	157	194	212	164	153	116	99	102	80	66	
All-Sky Solar Radiation	RadAvg	1.52	2.74	4.35	5.18	5.08	5.61	5.13	4.44	3.48	2.20	1.38	1.15	
	RadStd	0.06	0.09	0.19	0.38	0.45	0.55	0.52	0.41	0.28	0.17	0.12	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page